ENVS 220 – Applied Environmental Analysis

Condensed Syllabus (last updated: Spring 2026)

Description

The analysis of quantitative data remains essential for addressing a wide range of local and national environmental issues. Data can be used to monitor resources, track changes over time, and advocate for policy and behavioral change. In this course, we will explore the processes used to understand, analyze, and solve environmental problems through a quantitative lens. Students will be introduced to mathematical and statistical concepts, along with statistical software tools like R and STELLA. This course emphasizes both applied static and dynamic modeling of environmental systems.

Logistics

- Lecture: One ~1.5 hours session per week
- Practical: One ~1.5 hours session per week

Materials

No required textbooks. Course is based on:

- Reimann C., Filzmoser P., Garrett R., Dutter R. *Statistical Data Analysis Explained: Applied Environmental Statistics with R*. John Wiley & Sons. 2008
- Greg Langkamp and Joseph Hull. *Quantitative Reasoning and the Environment: Mathematical Modeling in Context*. Pearson Publishing. 2007 (freely available online)

Schedule

Week 1) Course Logistics
Week 2) Descriptive Statistics (Introduction to R)
Week 3) Data Distributions (Graphing and Hypothesis Testing)
Week 4) Analyzing Relationships (Two Variable Analysis)
Week 5) Multivariate Relationships (Regressions)
Week 6) Midterm I
Week 7) Principal Component Analysis (PCA)
Week 8) Linear and Exponential Functions (Introduction to STELLA)
Week 9) Complex Functions (Mono Lake Project)
Week 10) Difference Equations (Modeling Population Trends)
Week 11) Logistic Growth (Advanced Modeling Techniques)
Week 12) Midterm II
Week 13) Final Preparations
Week 14) Project Presentations

Evaluation

- Attendance (5%)
- Nine Homework (36%)
- Nine In-class Exercise (9%)
- Two Midterms (30%)
- Final Project Presentation and Report (20%)